

Adaptive Feature-Weighted 1D-CNN with Emotion-Aware Augmentation for Robust Speech Emotion Recognition

Project Proposal Document

Domain: Speech Processing & Affective Computing | Base Paper: Chourasia et al.,
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1. Problem Statement

Speech Emotion Recognition (SER) aims to automatically identify a speaker's emotional state from the acoustic properties of their voice. It is a core enabling technology for human–computer interaction, mental-health assessment, healthcare monitoring, intelligent virtual assistants, call-centre analytics, and human–robot interaction. Despite substantial progress, building an SER system that is simultaneously **accurate, generalizable, and computationally lightweight** remains an open problem.

State-of-the-art deep architectures such as transformers (Wav2Vec2, Audio Spectrogram Transformer) and hybrid CNN–LSTM networks achieve strong accuracy but demand large-scale training data and heavy computational resources, making them unsuitable for real-time or resource-constrained deployment (edge devices, embedded systems, mobile applications). Conversely, lightweight models built on handcrafted acoustic features typically make two simplifying assumptions that limit their performance: (i) all extracted acoustic features (MFCC, ZCR, RMSE) are treated as **equally important for every utterance**, even though the discriminative value of each feature varies from sample to sample and from emotion to emotion; and (ii) **identical data augmentation is applied uniformly to all emotion classes**, ignoring that each emotion has distinct natural speech characteristics (pitch range, speaking rate, energy), which can distort the emotional content of augmented samples.

In addition, existing lightweight models rely on **single-scale convolutional kernels** that cannot simultaneously capture short-term acoustic events (energy bursts, micro-pitch variations) and longer-term prosodic patterns (intonation contours, speaking rate), and they employ **standard cross-entropy loss**, which does not explicitly penalize the systematic confusion observed between acoustically similar emotion pairs such as *sad–neutral* and *angry–fear*.

Therefore, the problem addressed by this project is: to design a Conv1D-based speech emotion recognition framework that retains the computational efficiency of lightweight feature-based models while overcoming the above limitations through adaptive feature weighting, emotion-aware augmentation, multi-scale temporal feature extraction, and confusion-aware discriminative learning — evaluated on a combined multi-corpus dataset (RAVDESS, SAVEE, TESS, CREMA-D) across seven emotional categories.

2. Literature Survey

2.1 Base Paper

Chourasia, N., Lamba, C. S., & Gupta, A. K. (2026). “A 1D-CNN with advanced data augmentation for robust speech emotion recognition.” *Scientific Reports*. The base paper proposes a lightweight one-dimensional convolutional neural network (1D-CNN) for SER. Three complementary handcrafted acoustic features are extracted frame-wise from each 3-second utterance — 40 Mel-Frequency Cepstral

Coefficients (MFCC) capturing the spectral envelope, Zero-Crossing Rate (ZCR) capturing temporal signal transitions, and Root Mean Square Energy (RMSE) capturing loudness dynamics — and concatenated into a unified sequential vector of shape (2376, 1). The network consists of five stacked Conv1D blocks (512→512→256→256→128 filters), each followed by batch normalization and max-pooling, then dense layers and a 7-way Softmax classifier (≈ 7.19 M parameters). Four benchmark corpora — RAVDESS, SAVEE, TESS, and CREMA-D — are fused into a combined dataset of 12,162 samples (72:8:20 train/validation/test split), and four augmentation techniques (Gaussian noise injection, time shifting, pitch modification, time stretching) are applied only to the training subset. The model achieves **94.91% test accuracy** and a **Macro F1-score of 0.94**, with strong statistical reliability (MCC 0.9294, Cohen's Kappa 0.9292, AUC 0.9963), demonstrating that a well-designed lightweight CNN with discriminative features can rival far heavier architectures.

2.2 Related Work

Research in SER has evolved along three major directions: (i) handcrafted features with classical machine learning, (ii) end-to-end deep learning architectures, and (iii) augmentation-based strategies for data scarcity and class imbalance. Representative studies are summarized in Table 1.

Study	Method / Features	Dataset(s)	Reported Result	Limitation
Krishna et al. (2022) [18]	MFCC, MEL, Chroma + SVM / MLP	SER speech datasets	Improved accuracy with hybrid features	Requires manual feature engineering; limited representation power
Badshah et al. (2017) [19]	Spectrograms + CNN with transfer learning (AlexNet)	Berlin EMO-DB	High accuracy for most emotions	Poor performance on some emotions (fear); small dataset
Meng et al. (2019) [20]	ADRRN: dilated CNN + residual + BiLSTM + attention	IEMOCAP, EMO-DB	63.84% accuracy	Complex architecture; high computational cost
Zhao et al. (2019) [23]	Hybrid 1D/2D CNN–LSTM	RAVDESS	90.2% accuracy, F1 0.90	Increased model complexity; single-dataset evaluation
Pepino et al. (2021) [24]	Wav2Vec2 transformer embeddings	Multi-dataset	92.0% accuracy, strong generalization	Needs large pretrained models; high computational cost
Kilimci et al. (2025) [26]	End-to-end CNN on raw audio	SER benchmarks	Up to 99.46% accuracy	Overfitting risk; dependent on dataset characteristics
Ahmed et al. (2023) [28]	Ensemble 1D-CNN–LSTM–GRU + augmentation	Multi-dataset (multilingual)	93.5% accuracy, F1 0.93	High training complexity and resource demands
Saloumi et al. (2023) [29]	MFCC + 1D-CNN with augmentation	RAVDESS	83% accuracy	Small dataset; sensitive to data diversity; MFCC-only
Baek & Lee (2023) [4]	DCGAN-based synthetic augmentation + CNN	SER datasets	Improved robustness	GAN training instability; synthetic data reliability concerns
Dasude et al. (2024) [43]	CNN, LSTM, ConvLSTM comparison	TESS+RAVDESS+S AVEE+CREMA-D	50.6% on combined corpus	Moderate accuracy under dataset heterogeneity

Study	Method / Features	Dataset(s)	Reported Result	Limitation
Akinpelu et al. (2024) [66]	Vision Transformer (ViT)	RAVDESS	93.8% accuracy	Large data requirements; heavy architecture
Bhanbhro et al. (2025) [67]	Attention-enhanced CNN-LSTM	RAVDESS	94.2% accuracy	Higher complexity; single-corpus evaluation
Tang et al. (2025) [68]	CNN-Transformer + multidimensional attention	MELD, RAVDESS	95.2% accuracy (highest)	Significantly larger model; heavy computational resources
Base paper — Chourasia et al. (2026)	MFCC+ZCR+RMSE + 5-block 1D-CNN + uniform augmentation	RAVDESS+SAVEE+ TESS+CREMA-D	94.91% accuracy, Macro F1 0.94	Equal feature treatment; uniform augmentation; single-scale kernels; plain cross-entropy loss

Table 1: Comparative summary of representative SER studies.

Collectively, the literature shows a clear trade-off: transformer- and attention-based models (Tang et al., Bhanbhro et al., Akinpelu et al.) push accuracy upward but at substantial computational expense, while lightweight feature-based models (Saloumi et al., base paper) remain deployable but leave measurable performance on the table due to static feature fusion and uniform augmentation. This project targets precisely that gap.

3. Limitations of Existing Approaches

L1 — Equal (static) feature treatment. Existing lightweight SER pipelines, including the base paper, simply concatenate MFCC, ZCR, and RMSE, implicitly assuming each feature contributes equally to every prediction. In reality, some utterances are tone-distinctive (MFCC-dominant) while others are energy-distinctive (RMSE-dominant); static concatenation prevents the network from exploiting this per-sample variability.

L2 — Emotion-agnostic, uniform augmentation. The same four augmentation techniques are applied identically to all seven emotion classes. Because each emotion has characteristic acoustics (happy: wide pitch variation; sad: slow rate and low energy; angry: high energy; fear: erratic fluctuations), uniform augmentation can push augmented samples away from the natural acoustic manifold of their class, injecting label noise instead of useful diversity.

L3 — Single-scale temporal receptive fields. Each convolutional layer in existing 1D-CNN pipelines uses a single kernel size, so any one layer observes emotional cues at only one temporal resolution. Short-lived events (energy bursts, micro-pitch variations) and slower prosodic structures (intonation contours, stress patterns) are therefore never captured in parallel at the same depth.

L4 — Confusion-blind training objective. Models are trained with standard categorical cross-entropy, which treats all misclassifications as equally costly. Yet confusion matrices consistently show that errors concentrate in acoustically similar pairs — sad↔neutral (both low-energy, slow) and angry↔fear (both high-arousal, intense). Nothing in the objective explicitly encourages the model to separate these hard pairs.

L5 — Heavy alternatives are impractical for deployment. Transformer, ViT, and hybrid recurrent architectures that could address the above issues require large pretrained models, large-scale datasets, and high computational resources, contradicting the goal of real-time SER on edge and embedded devices.

L6 — Limited evaluation practice. Several existing studies evaluate on a single corpus (often RAVDESS), overstating generalization; ablation analyses of individual features and augmentations are usually missing, so the contribution of each component remains unquantified.

4. Novelty of the Proposed Work

This project retains the efficient Conv1D backbone of the base paper and introduces **four lightweight, mutually complementary novelties**. Each novelty directly answers one of the limitations identified in Section 3, and each can be enabled or disabled independently, giving the project a natural ablation-study structure that the base paper itself lacked.

Novelty 1 — Adaptive Feature Weighting (AFW) [addresses L1]

Instead of concatenating MFCC, RMSE, and ZCR with fixed equal importance, AFW inserts a small attention-style gating module between feature extraction and the Conv1D backbone. The module takes the three feature streams, computes a per-sample importance score for each (compact dense layer followed by a softmax), and scales every stream by its learned weight before fusion. A tone-distinctive utterance may thus receive weights of approximately 60% MFCC / 25% RMSE / 15% ZCR, while an energy-distinctive utterance shifts emphasis toward RMSE. The weights are trained jointly with the network, add only a few thousand parameters, and preserve the lightweight design goal.

Novelty 2 — Emotion-Aware Adaptive Augmentation (EAAA) [addresses L2]

EAAA replaces uniform augmentation with an emotion-conditioned augmentation policy: the technique applied to each training sample is selected to mirror the natural acoustic variability of its emotion class, so augmented data stays faithful to how that emotion actually sounds. The mapping, grounded in the speech-emotion literature and in the base paper's own spectrogram analysis, is given in Table 2.

Emotion	Natural Acoustic Characteristic	Assigned Augmentation
Happy	Wide pitch variation; expressive, rhythmic delivery	Pitch shifting (± 2 semitones)
Sad	Slow speaking rate; low, smooth energy	Time stretching (0.8–1.2 $\times$)
Angry	High energy, vocal strain, intense peaks	Noise injection (0.001–0.005)
Fear	Erratic fluctuations; unstable low-to-mid energy	Time shifting (± 0.2 s) + light noise
Surprise	Sudden bursts of spectral energy; brief utterances	Time shifting (± 0.2 s)
Disgust	Controlled, tense, low-frequency dominance	Mild pitch shift + mild stretch
Neutral	Steady, evenly distributed spectral energy	Light noise injection only

Table 2: Emotion-to-augmentation mapping used by EAAA (all seven classes).

Novelty 3 — Multi-Scale Temporal Convolution (MSTC) [addresses L3]

The first convolutional block is replaced by three parallel Conv1D branches with kernel sizes 3, 5, and 7 operating on the same input, whose outputs are concatenated channel-wise before entering the remaining backbone. The small kernel captures rapid, short-term acoustic events (energy bursts, micro-pitch variation), the medium kernel captures phoneme-level spectral transitions, and the large kernel captures slower prosodic structure (intonation contours, speaking-rate patterns) — all at the same network depth. Because the total filter budget of the block is split across the three branches rather than tripled, the parameter count remains close to the original single-scale block, preserving computational efficiency.

Novelty 4 — Confusion-Aware Discriminative Loss (CADL) [addresses L4]

Standard categorical cross-entropy is replaced by a composite objective: cross-entropy combined with a focal-style modulation and an additional pairwise penalty term applied to the empirically confusable emotion pairs (sad $\leftrightarrow$ neutral and angry $\leftrightarrow$ fear). The focal component down-weights easy, already-correct samples so training concentrates on hard examples, while the pairwise penalty adds an extra margin cost whenever a sample is misclassified into its known confusion partner. This directly targets the dominant error mode reported in the base paper's confusion matrix without adding any inference-time cost — the network architecture and prediction pipeline are unchanged; only training is affected.

Novelty	Limitation Addressed	Where It Acts	Cost
N1: Adaptive Feature Weighting (AFW)	L1 — equal feature treatment	Feature-fusion level	Few thousand extra parameters (<0.1%)
N2: Emotion-Aware Adaptive Augmentation (EAAA)	L2 — uniform augmentation	Data level (training only)	Zero model overhead; rule-based selection
N3: Multi-Scale Temporal Convolution (MSTC)	L3 — single-scale kernels	Architecture level (first block)	$\approx$ parameter-neutral (filter budget split)
N4: Confusion-Aware Discriminative Loss (CADL)	L4 — confusion-blind objective	Training-objective level	Zero inference-time overhead

Table 3: Summary of the four proposed novelties.

5. Proposed Solution

The proposed system is an **Adaptive Feature-Weighted, Multi-Scale 1D-CNN with Emotion-Aware Augmentation and Confusion-Aware Training**. It follows the base paper's efficient pipeline end-to-end, upgrading four specific stages.

5.1 Dataset and Preprocessing

Four public benchmark corpora are fused into a single multi-speaker dataset: RAVDESS (1,440 speech samples, 24 actors), TESS (2,800 samples, 2 female speakers), SAVEE (480 samples, male speakers), and CREMA-D (7,442 samples, 91 actors) — 12,162 samples in total, mapped to seven emotion classes (disgust, happy, sad, neutral, fear, angry, surprise). All audio is resampled to 22,050 Hz, amplitude-normalized to $[-1, 1]$, and zero-padded or truncated to a fixed 3-second duration. The data is split 72:8:20 into training (8,756), validation (973), and test (2,433) subsets **before** augmentation, guaranteeing that validation and test sets remain untouched and preventing data leakage.

5.2 Emotion-Aware Augmentation Stage (EAAA)

Augmentation is applied only to the training subset. For each training sample, the augmentation technique is selected according to the emotion-to-augmentation policy of Table 2, expanding the effective training set (to approximately 11,700 samples, matching the base paper's expansion budget for a fair comparison) while keeping every augmented sample acoustically consistent with its emotion class.

5.3 Feature Extraction and Adaptive Weighting (AFW)

For every 3-second sample, frame-level features are computed with a 25 ms window and 10 ms hop: 40 MFCC coefficients, ZCR, and RMSE. Before fusion, the AFW module computes per-sample softmax importance weights over the three streams and rescales each stream accordingly; the weighted streams are then concatenated into the sequential input tensor of shape (2376, 1) expected by the backbone. The learned weights are also interpretable — they can be visualized per emotion to show which acoustic cue drives each class.

5.4 Multi-Scale Conv1D Backbone (MSTC)

The backbone follows the base architecture — five convolutional stages with batch normalization and max-pooling (pool size 2), filters decreasing $512 \rightarrow 512 \rightarrow 256 \rightarrow 256 \rightarrow 128$ — with the first stage replaced by the multi-scale block: three parallel Conv1D branches (kernel sizes 3, 5, 7) whose outputs are concatenated channel-wise. The final feature maps are flattened and passed through a Dense(512) layer with ReLU and batch normalization, ending in a 7-neuron Softmax output. Total parameters remain close to the base model's ≈ 7.19 M, keeping the model deployable on resource-constrained hardware.

5.5 Confusion-Aware Training (CADL)

The network is trained with the Adam optimizer (learning rate 0.001, batch size 32, 50 epochs) using the composite CADL objective: categorical cross-entropy with focal modulation plus the pairwise margin penalty on the sad $\leftrightarrow$ neutral and angry $\leftrightarrow$ fear pairs. Early stopping on validation loss guards against overfitting.

5.6 Evaluation Protocol

Performance is measured on the held-out test set using accuracy, per-class and macro precision/recall/F1, specificity, G-Mean, MCC, Cohen's Kappa, AUC, confusion matrices, and a 95% confidence interval for accuracy — the same metric suite as the base paper, enabling direct

comparison. In addition, a component-wise **ablation study** is conducted (base model, +AFW, +EAAA, +MSTC, +CADL, and all combined) to quantify the individual contribution of each novelty — an analysis explicitly missing from the base paper.

5.7 Workflow Summary

Step	Stage	Description
1	Data fusion	Merge RAVDESS + SAVEE + TESS + CREMA-D; map to 7 emotion classes (12,162 samples)
2	Preprocessing	Resample 22,050 Hz → normalize $[-1,1]$ → pad/truncate to 3 s
3	Split	72:8:20 train / validation / test split (before augmentation)
4	EAAA (Novelty 2)	Emotion-conditioned augmentation applied to training data only
5	Feature extraction	40 MFCC + ZCR + RMSE per frame (25 ms window, 10 ms hop)
6	AFW (Novelty 1)	Per-sample learned weighting of the three feature streams before fusion
7	MSTC backbone (Novelty 3)	Multi-scale first block (kernels 3/5/7) + 4 further Conv1D stages
8	CADL training (Novelty 4)	Focal cross-entropy + pairwise confusion penalty; Adam, lr 0.001
9	Evaluation	Full metric suite + confusion matrix + 4-way ablation study

Table 4: End-to-end workflow of the proposed framework.

6. Expected Outcome

Improved recognition accuracy over the base paper. By resolving static feature fusion, uniform augmentation, single-scale receptive fields, and the confusion-blind objective simultaneously, the framework is expected to exceed the base paper's 94.91% test accuracy and 0.94 Macro F1 on the identical combined-corpus test protocol, with a target in the range of 95.5–96.5%.

Reduced confusion between hard emotion pairs. CADL is expected to measurably lower the sad↔neutral and angry↔fear misclassification counts in the confusion matrix — the dominant error mode of the base model — improving per-class F1 for sad and neutral (currently the weakest at 0.93).

More realistic, class-faithful augmented data. EAAA is expected to raise the quality of augmentation, particularly benefiting minority classes such as surprise (absent from CREMA-D), improving robustness without injecting label noise.

Interpretability of acoustic cues. The AFW module yields per-sample and per-class feature-importance weights, providing insight into which acoustic dimension (spectral, temporal, or energy) drives each emotion — a form of built-in explainability absent from the base model.

Preserved computational efficiency. All four novelties are lightweight: total parameters remain approximately at the base model's 7.19 M, and inference cost is unchanged (CADL and EAAA affect training only), keeping the system suitable for real-time and edge deployment.

A complete ablation analysis. The project will deliver the component-wise ablation (base, +AFW, +EAAA, +MSTC, +CADL, all combined) that the base paper listed as missing future work, quantifying each novelty's individual contribution.

Deliverables. A trained SER model, full source code, evaluation reports (metric tables, training curves, confusion matrices, ablation tables), and a project report suitable for academic submission.

7. Key References

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